

Unit 35: Inverse Trigonometric Functions

1. The idea, in plain words

You already know how to find $\sin(30^\circ)$ — but what about the reverse question: “what angle has a sine of $\frac{1}{2}$?” That’s exactly what inverse trig functions answer. $\arcsin x$ (also written $\sin^{-1} x$) asks “what angle has this sine value?” and hands back an angle.

The one tricky wrinkle: since sine, cosine, and tangent repeat their values over and over (infinitely many angles share the same sine), each inverse trig function has to agree on picking just one specific angle as “the” answer, out of all the possibilities. Each one restricts its output to a narrow window, called its range:

- $\arcsin x$ always gives an angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.
- $\arccos x$ always gives an angle in $[0, \pi]$.
- $\arctan x$ always gives an angle in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

A powerful visual trick for evaluating compositions like $\sin(\arccos x)$: picture $\theta = \arccos x$ as an actual angle in a right triangle. Since $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$, if $\cos \theta = x$, you can draw a triangle with adjacent side x and hypotenuse 1 — then the Pythagorean theorem hands you the missing side, and you can read off whatever other trig ratio you need (like $\sin \theta$) directly from the triangle.

Derivatives of the inverse trig functions follow their own distinctive patterns — different enough from anything before that they’re worth memorizing as a fresh set of formulas, but once memorized, they combine with the chain rule exactly like everything else you’ve learned.

2. Toolbox

Standard values worth memorizing (from the familiar 30° - 45° - 60° angles):

$$\begin{aligned} \arcsin\left(\frac{1}{2}\right) &= \frac{\pi}{6} & \arcsin\left(\frac{\sqrt{2}}{2}\right) &= \frac{\pi}{4} & \arcsin\left(\frac{\sqrt{3}}{2}\right) &= \frac{\pi}{3} \\ \arccos(0) &= \frac{\pi}{2} & \arccos\left(\frac{1}{2}\right) &= \frac{\pi}{3} & \arctan(1) &= \frac{\pi}{4} \end{aligned}$$

The right-triangle technique for compositions: to evaluate something like $\sin(\arccos x)$ or $\tan(\arcsin x)$, draw a right triangle where the given ratio matches the inverse function’s definition, use the Pythagorean theorem to find the missing side, then read off the requested ratio.

Derivatives:

$$\frac{d}{dx}[\arcsin x] = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx}[\arccos x] = \frac{-1}{\sqrt{1-x^2}} \quad \frac{d}{dx}[\arctan x] = \frac{1}{1+x^2}$$

Chain rule versions (for u instead of plain x):

$$\frac{d}{dx}[\arcsin u] = \frac{u'}{\sqrt{1-u^2}} \quad \frac{d}{dx}[\arccos u] = \frac{-u'}{\sqrt{1-u^2}} \quad \frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$$

3. Common mistakes

- Forgetting the restricted range. $\arcsin x$ never returns an angle outside $[-\frac{\pi}{2}, \frac{\pi}{2}]$, even though sine itself repeats — always double-check your answer actually falls in the correct window.
- Sign errors when reading a triangle for a negative input. If $\arcsin x$ has a negative input, the angle is negative (in the fourth-quadrant-like part of its range) — track carefully which sides of the triangle should be positive or negative based on the specific inverse function's range.
- Mixing up the three derivative formulas, especially the sign difference between $\arcsin$ and $\arccos$, and the “ $1+x^2$ ” pattern unique to $\arctan$ (no square root involved there, unlike the other two).
- Forgetting the chain rule multiplier u' when the input isn't plain x .
- Forgetting the domain restriction on $\arcsin$ and $\arccos$ — their inputs must stay within $[-1, 1]$.

4. Problem Set

Warm-up

1. Find $\arcsin\left(\frac{1}{2}\right)$.
2. Find $\arccos(0)$.
3. Find $\arctan(1)$.
4. Find $\arcsin\left(-\frac{\sqrt{2}}{2}\right)$.
5. Find $\arccos(1)$.
6. State the formula for $\frac{d}{dx}[\arcsin x]$.
7. State the formula for $\frac{d}{dx}[\arctan x]$.

Standard

8. Find $\sin\left(\arccos\left(\frac{3}{5}\right)\right)$.
9. Find $\cos\left(\arcsin\left(\frac{5}{13}\right)\right)$.
10. Find $\tan\left(\arcsin\left(\frac{1}{3}\right)\right)$.
11. Find $\frac{d}{dx}[\arcsin(2x)]$.
12. Find $\frac{d}{dx}[\arctan(x^2)]$.
13. Find $\frac{d}{dx}[\arccos(3x)]$.
14. Find $\sec(\arctan(2))$.

Challenge

15. $y = \arctan(\sin x)$. Find $\frac{dy}{dx}$.
16. Find $\sin\left(\arccos\left(\frac{\sqrt{2}}{2}\right)\right)$.
17. Find $\tan\left(\arcsin\left(-\frac{1}{2}\right)\right)$.
18. Find $\frac{d}{dx}\left[x \arcsin(x) + \sqrt{1-x^2}\right]$.

19. Find $\cos(\arctan(\frac{3}{4}))$.

Applied

20. A ladder leans against a wall, reaching 15 ft up with its base 8 ft from the wall. Find the angle of elevation $\theta = \arctan\left(\frac{15}{8}\right)$, in degrees.
21. A camera tracks an object's horizontal distance $x(t)$ (in feet) from a fixed point, with the tracking angle given by $\theta(t) = \arctan\left(\frac{x(t)}{50}\right)$. At the moment $x = 30$ ft and $\frac{dx}{dt} = 10$ ft/s, find $\frac{d\theta}{dt}$.
22. A viewing angle is modeled by $\theta = \arcsin\left(\frac{x}{10}\right)$. Find $\frac{d\theta}{dx}$, then evaluate it at $x = 6$.
23. A ship has traveled 3 miles east and is currently 5 miles from its starting point. Its bearing angle from the starting direction is $\theta = \arccos\left(\frac{3}{5}\right)$. Find θ in degrees.